Smart Healthcare Network Design

CSE421 - Computer Networks

Lab Project Documentation

Student ID: 22101783

Base Network: 17.83.0.0/16

April 2026

1. VLSM Subnetting

1.1 Base Network Calculation

Student ID: 22101783

Last 4 digits: 1783

Split into two parts: 17 and 83

Base network: 17.83.0.0/16

1.2 VLSM Allocation Table

Subnets are allocated in the required order: CH (/23), DC (/24), RL (/25), PH (/25), EU (/25), RC (/26).

Unit	CIDR	Network Addr	Broadcast	Gateway	Subnet Mask	Usable	Needed
CH	/23	17.83.0.0	17.83.1.255	17.83.0.1	255.255.254.0	510	300
DC	/24	17.83.2.0	17.83.2.255	17.83.2.1	255.255.255.0	254	150
RL	/25	17.83.3.0	17.83.3.127	17.83.3.1	255.255.255.128	126	120
PH	/25	17.83.3.128	17.83.3.255	17.83.3.129	255.255.255.128	126	100
EU	/25	17.83.4.0	17.83.4.127	17.83.4.1	255.255.255.128	126	80
RC	/26	17.83.4.128	17.83.4.191	17.83.4.129	255.255.255.192	62	60

1.3 VLSM Tree

17.83.0.0/16 (Base Network)

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+-- 17.83.0.0/23 ..... CH (300 devices, 510 usable hosts)

|

+-- 17.83.2.0/24 ..... DC (150 devices, 254 usable hosts)

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+-- 17.83.3.0/25 ..... RL (120 devices, 126 usable hosts)

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+-- 17.83.3.128/25 .... PH (100 devices, 126 usable hosts)

|

+-- 17.83.4.0/25 ..... EU (80 devices, 126 usable hosts)

|

+-- 17.83.4.128/26 .... RC (60 devices, 62 usable hosts)

|

+-- Point-to-Point Links:

  +-- 17.83.4.192/30 .... CH-DC Link

  +-- 17.83.4.196/30 .... CH-RL Link

  +-- 17.83.4.200/30 .... CH-EU Link

  +-- 17.83.4.204/29 .... Core-Switch (RL, PH, EU)

  +-- 17.83.4.212/30 .... PH-RC Link

  +-- 17.83.4.216/30 .... EU-RC Link

1.4 Point-to-Point Links

Link	Subnet	End A	End B
CH-DC Link	17.83.4.192/30	17.83.4.193	17.83.4.194
CH-RL Link	17.83.4.196/30	17.83.4.197	17.83.4.198
CH-EU Link	17.83.4.200/30	17.83.4.201	17.83.4.202
Core-Switch	17.83.4.204/29	RL: .205	PH: .206, EU: .207
PH-RC Link	17.83.4.212/30	17.83.4.213	17.83.4.214
EU-RC Link	17.83.4.216/30	17.83.4.217	17.83.4.218

2. IP Addressing Table

Complete IP addressing for all devices in the Smart Healthcare Network.

Device	Interface	IP Address	Subnet Mask	Subnet	Method
CH-Router	Gig0/0 (to DC)	17.83.4.193	255.255.255.252	17.83.4.192/30	Static
CH-Router	Gig0/1 (to RL)	17.83.4.197	255.255.255.252	17.83.4.196/30	Static
CH-Router	Gig0/2 (to EU)	17.83.4.201	255.255.255.252	17.83.4.200/30	Static
CH-Router	Vlan1 (LAN)	17.83.0.1	255.255.254.0	17.83.0.0/23	Static
CH-PC1	Fa0	17.83.0.2	255.255.254.0	17.83.0.0/23	Static
CH-PC2	Fa0	17.83.0.3	255.255.254.0	17.83.0.0/23	Static
CH-DNS-Web	Fa0	17.83.0.4	255.255.254.0	17.83.0.0/23	Static
CH-Email	Fa0	17.83.0.5	255.255.254.0	17.83.0.0/23	Static
DC-Router	Gig0/0 (to CH)	17.83.4.194	255.255.255.252	17.83.4.192/30	Static
DC-Router	Gig0/1 (LAN)	17.83.2.1	255.255.255.0	17.83.2.0/24	Static
DC-PC1	Fa0	DHCP	255.255.255.0	17.83.2.0/24	DHCP
DC-PC2	Fa0	DHCP	255.255.255.0	17.83.2.0/24	DHCP
DC-Email	Fa0	DHCP/Static	255.255.255.0	17.83.2.0/24	DHCP/ Static
RL-Router	Gig0/0 (to CH)	17.83.4.198	255.255.255.252	17.83.4.196/30	Static
RL-Router	Gig0/1 (Core-SW)	17.83.4.205	255.255.255.248	17.83.4.204/29	Static
RL-Router	Gig0/2 (LAN)	17.83.3.1	255.255.255.128	17.83.3.0/25	Static
RL-PC1	Fa0	DHCP (17.83.3.13)	255.255.255.128	17.83.3.0/25	DHCP
RL-PC2	Fa0	DHCP (17.83.3.14)	255.255.255.128	17.83.3.0/25	DHCP
RL-DHCP Server	Fa0	17.83.3.2	255.255.255.128	17.83.3.0/25	Static
RL-Email	Fa0	17.83.3.12	255.255.255.128	17.83.3.0/25	Static
PH-Router	Gig0/0 (Core-SW)	17.83.4.206	255.255.255.248	17.83.4.204/29	Static
PH-Router	Gig0/1 (to RC)	17.83.4.213	255.255.255.252	17.83.4.212/30	Static
PH-Router	Gig0/2 (LAN)	17.83.3.129	255.255.255.128	17.83.3.128/25	Static

PH-PC1	Fa0	DHCP (17.83.3.132)	255.255.255.128	17.83.3.128/25	DHCP
PH-PC2	Fa0	DHCP (17.83.3.133)	255.255.255.128	17.83.3.128/25	DHCP
PH-DHCP Server	Fa0	17.83.3.130	255.255.255.128	17.83.3.128/25	Static
PH-Email	Fa0	DHCP/Static	255.255.255.128	17.83.3.128/25	DHCP/ Static
EU-Router	Gig0/0 (to CH)	17.83.4.202	255.255.255.252	17.83.4.200/30	Static
EU-Router	Gig0/1 (Core-SW)	17.83.4.207	255.255.255.248	17.83.4.204/29	Static
EU-Router	Gig0/2 (to RC)	17.83.4.217	255.255.255.252	17.83.4.216/30	Static
EU-Router	Vlan1 (LAN)	17.83.4.1	255.255.255.128	17.83.4.0/25	Static
EU-PC1	Fa0	DHCP	255.255.255.128	17.83.4.0/25	DHCP
EU-PC2	Fa0	DHCP	255.255.255.128	17.83.4.0/25	DHCP
EU-Email	Fa0	DHCP/Static	255.255.255.128	17.83.4.0/25	DHCP/ Static
RC-Router	Gig0/0 (to PH)	17.83.4.214	255.255.255.252	17.83.4.212/30	Static
RC-Router	Gig0/1 (to EU)	17.83.4.218	255.255.255.252	17.83.4.216/30	Static
RC-Router	Gig0/2 (LAN)	17.83.4.129	255.255.255.192	17.83.4.128/26	Static
RC-PC1	Fa0	17.83.4.130	255.255.255.192	17.83.4.128/26	Static
RC-PC2	Fa0	17.83.4.131	255.255.255.192	17.83.4.128/26	Static
RC-Email	Fa0	17.83.4.132	255.255.255.192	17.83.4.128/26	Static

3. DHCP Configuration

DC and EU use router-based DHCP. RL and PH use dedicated DHCP servers. CH and RC use static IP addressing.

Device	Type	Pool Name	Range	Gateway	DNS
DC-Router	Router DHCP	DC-POOL	17.83.2.0/24	17.83.2.1	17.83.0.4
EU-Router	Router DHCP	EU-POOL	17.83.4.0/25	17.83.4.1	17.83.0.4
RL-DHCP Server	Dedicated Server	RL-POOL	17.83.3.13+ (/25)	17.83.3.1	17.83.0.4
PH-DHCP Server	Dedicated Server	serverPool	17.83.3.130+ (/25)	17.83.3.129	17.83.0.4

4. Server Configuration

Server	IP Address	Services	Configuration Details
CH-DNS-Web	17.83.0.4	DNS + HTTP	www.health.local = 17.83.0.4, ch.health.com = 17.83.0.5, rl.health.com = 17.83.3.12
CH-Email	17.83.0.5	SMTP + POP3	Domain: ch.health.com, User: admin/admin123
DC-Email	DHCP/Static	SMTP + POP3	Domain: dc.health.com, User: admin/admin123
RL-DHCP	17.83.3.2	DHCP	Pool: 17.83.3.13+, GW: 17.83.3.1, DNS: 17.83.0.4
RL-Email	17.83.3.12	SMTP + POP3	Domain: rl.health.com, User: admin/admin123
PH-DHCP	17.83.3.130	DHCP	Pool: 17.83.3.140+, GW: 17.83.3.129, DNS: 17.83.0.4
PH-Email	DHCP/Static	SMTP + POP3	Domain: ph.health.com, User: admin/admin123
EU-Email	DHCP/Static	SMTP + POP3	Domain: eu.health.com, User: admin/admin123
RC-Email	17.83.4.132	SMTP + POP3	Domain: rc.health.com, User: admin/admin123

5. Routing Configuration

5.1 Routing Summary

Router	Routing Type	Commands
CH-Router	Next-hop static	ip route 17.83.2.0 255.255.255.0 17.83.4.194
CH-Router	RIPv2 + redistribute	router rip / version 2 / network 17.0.0.0 / redistribute static / no auto-summary
DC-Router	Default static	ip route 0.0.0.0 0.0.0.0 17.83.4.193
EU-Router	Exit-interface static	ip route 17.83.0.0 255.255.254.0 GigabitEthernet0/0
EU-Router	RIPv2	router rip / version 2 / network 17.0.0.0 / no auto-summary
RL-Router	RIPv2	router rip / version 2 / network 17.0.0.0 / no auto-summary
PH-Router	Recursive static	ip route 17.83.2.0 255.255.255.0 17.83.4.205
PH-Router	RIPv2	router rip / version 2 / network 17.0.0.0 / no auto-summary
RC-Router	Floating static (AD=130)	ip route 17.83.0.0 255.255.254.0 17.83.4.217 130
RC-Router	RIPv2	router rip / version 2 / network 17.0.0.0 / no auto-summary

5.2 Detailed Router Commands

CH-Router (Central Hospital)

```
enable
configure terminal
! --- Interface Configuration ---
interface GigabitEthernet0/0
  ip address 17.83.4.193 255.255.255.252
  no shutdown
interface GigabitEthernet0/1
  ip address 17.83.4.197 255.255.255.252
  no shutdown
interface GigabitEthernet0/2
  ip address 17.83.4.201 255.255.255.252
  no shutdown
interface Vlan1
  ip address 17.83.0.1 255.255.254.0
  no shutdown
! --- Static Route to DC (Next-Hop) ---
```

```
ip route 17.83.2.0 255.255.255.0 17.83.4.194
! --- RIPv2 with Static Redistribution ---
router rip
  version 2
  network 17.0.0.0
  redistribute static
  no auto-summary
end
```

DC-Router (Diagnostic Center)

```
enable
configure terminal
interface GigabitEthernet0/0
  ip address 17.83.4.194 255.255.255.252
  no shutdown
interface GigabitEthernet0/1
  ip address 17.83.2.1 255.255.255.0
  no shutdown
! --- Default Static Route to CH ---
ip route 0.0.0.0 0.0.0.0 17.83.4.193
! --- DHCP Pool ---
ip dhcp pool DC-POOL
  network 17.83.2.0 255.255.255.0
  default-router 17.83.2.1
  dns-server 17.83.0.4
end
```

RL-Router (Research Lab)

```
enable
configure terminal
interface GigabitEthernet0/0
  ip address 17.83.4.198 255.255.255.252
  no shutdown
interface GigabitEthernet0/1
```

```
 ip address 17.83.4.205 255.255.255.248
 no shutdown
interface GigabitEthernet0/2
 ip address 17.83.3.1 255.255.255.128
 no shutdown
! --- RIPv2 Only ---
router rip
 version 2
 network 17.0.0.0
 no auto-summary
end
```

PH-Router (Pharmacy Hub)

```
enable
configure terminal
interface GigabitEthernet0/0
 ip address 17.83.4.206 255.255.255.248
 no shutdown
interface GigabitEthernet0/1
 ip address 17.83.4.213 255.255.255.252
 no shutdown
interface GigabitEthernet0/2
 ip address 17.83.3.129 255.255.255.128
 no shutdown
! --- Recursive Static Route to DC via RL ---
ip route 17.83.2.0 255.255.255.0 17.83.4.205
! --- RIPv2 ---
router rip
 version 2
 network 17.0.0.0
 no auto-summary
end
```

EU-Router (Emergency Unit)

```
enable
configure terminal
interface GigabitEthernet0/0
 ip address 17.83.4.202 255.255.255.252
 no shutdown
interface GigabitEthernet0/1
 ip address 17.83.4.207 255.255.255.248
 no shutdown
interface GigabitEthernet0/2
 ip address 17.83.4.217 255.255.255.252
 no shutdown
interface Vlan1
 ip address 17.83.4.1 255.255.255.128
 no shutdown
! --- Exit-Interface Static Route to CH ---
ip route 17.83.0.0 255.255.254.0 GigabitEthernet0/0
! --- RIPv2 ---
router rip
 version 2
 network 17.0.0.0
 no auto-summary
! --- DHCP Pool ---
ip dhcp pool EU-POOL
 network 17.83.4.0 255.255.255.128
 default-router 17.83.4.1
 dns-server 17.83.0.4
end
```

RC-Router (Remote Clinic)

```
enable
configure terminal
interface GigabitEthernet0/0
 ip address 17.83.4.214 255.255.255.252
 no shutdown
```

```
interface GigabitEthernet0/1
  ip address 17.83.4.218 255.255.255.252
  no shutdown
interface GigabitEthernet0/2
  ip address 17.83.4.129 255.255.255.192
  no shutdown
! --- Floating Static Route to CH via EU (AD=130, backup) ---
ip route 17.83.0.0 255.255.254.0 17.83.4.217 130
! --- RIPv2 (primary path via PH) ---
router rip
  version 2
  network 17.0.0.0
  no auto-summary
end
```

6. Network Topology Description

The Smart Healthcare Network consists of 6 units interconnected as follows:

Central Hospital (CH) serves as the backbone of the network. It connects directly to the Diagnostic Center (DC) via a dedicated point-to-point /30 link. CH also connects directly to the Research Lab (RL) and the Emergency Unit (EU) through separate /30 links.

The Research Lab (RL), Pharmacy Hub (PH), and Emergency Unit (EU) are interconnected through a Core-Switch using a /29 subnet block (17.83.4.204/29), enabling dynamic routing among these three units via RIPv2.

The Remote Clinic (RC) connects to both PH and EU via separate /30 links, providing dual-path redundancy. The primary path to the network is via PH (learned through RIP), and a backup floating static route (AD=130) via EU activates automatically if the primary path fails.

Each unit has its own local switch connecting end devices (PCs, servers). CH-Router and EU-Router use HWIC-4ESW modules for built-in switch ports to serve their LANs.

DC is an isolated stub network connecting only to CH, with a default static route pointing back to CH for all outbound traffic.

7. Special Conditions and Redundancy

Condition 1: CH-RL Link Failure

If the direct link between CH and RL fails, traffic automatically reroutes through EU via the Core-Switch. This is handled by RIPv2 convergence; when CH-RL goes down, RIP recalculates and finds the alternative path CH -> EU -> Core-Switch -> RL.

Condition 2: RC Dual-Path Redundancy

RC maintains two paths to the rest of the network. The primary path is via PH, learned through RIPv2 (administrative distance 120). The backup path is a floating static route via EU with administrative distance 130. Since 120 < 130, the RIP route is preferred. If the PH path fails, the floating static route automatically takes over.

Condition 3: PH Recursive Static Fallback

PH has a recursive static route to reach DC (17.83.2.0/24) via RL's Core-Switch IP (17.83.4.205). This means PH relies on RL's routing table to reach DC. If the primary path from PH to CH via RL is unavailable, the recursive route provides an alternative by pointing through the Core-Switch.

8. Work Distribution

Task	Description	Assigned To
Topology Design	Placing routers, switches, PCs, servers; connecting all devices with correct cables	Member 1
VLSM Subnetting	Calculating all subnet addresses from base network 17.83.0.0/16 using VLSM	Member 2
Router IP Configuration	Assigning IP addresses to all router interfaces across all 6 units	Member 1
DHCP Configuration	Setting up DHCP pools on DC/EU routers and RL/PH dedicated servers	Member 3
Routing Configuration	Configuring static routes (next-hop, exit-interface, recursive, floating) and RIPv2	Member 2
Server Setup	Configuring DNS, Web, and Email servers with appropriate domains and DNS records	Member 3
Testing & Verification	End-to-end ping tests, email tests, web server access, redundancy testing	All Members
Documentation	VLSM tree, IP addressing table, router configs, topology diagram	All Members